

Lesson 3: Properties of Limits

Topic 1.5: Determining Limits Using Algebraic Properties of Limits

There will be times when you need to consider limits of functions symbolically, such as, $f(x)$, $g(x)$, and $h(x)$. When you are given combinations, compositions, and other modifications of these nonspecific functions, there are some important properties of limits that will prove essential to these general function types.

LIMIT PROPERTIES

Let b and c be real numbers, let n be a positive integer, and let f and g be functions.

1. Constant Rule:

$$\lim_{x \rightarrow c} k = k$$

2. Identity Rule:

$$\lim_{x \rightarrow c} x = c$$

3. Coefficient Rule:

$$\lim_{x \rightarrow c} [b \cdot f(x)] = b \cdot \lim_{x \rightarrow c} f(x)$$

4. Sum or Difference Rule:

$$\lim_{x \rightarrow c} [f(x) \pm g(x)] = \lim_{x \rightarrow c} f(x) \pm \lim_{x \rightarrow c} g(x)$$

5. Product Rule:

$$\lim_{x \rightarrow c} [f(x) \cdot g(x)] = \lim_{x \rightarrow c} f(x) \cdot \lim_{x \rightarrow c} g(x)$$

6. Quotient Rule:

$$\lim_{x \rightarrow c} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow c} f(x)}{\lim_{x \rightarrow c} g(x)} ; \lim_{x \rightarrow c} g(x) \neq 0$$

7. Power Rule:

$$\lim_{x \rightarrow c} [f(x)]^n = \left(\lim_{x \rightarrow c} f(x) \right)^n, n \text{ is rational}$$

8. Composite Function Rule:

$$\lim_{x \rightarrow c} f(g(x)) = f\left(\lim_{x \rightarrow c} g(x)\right)$$

9. Root Rule:

$$\lim_{x \rightarrow c} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow c} f(x)} \text{ if the root on the right side exists.}$$

Being skilled at reciting these rules in English will help you truly understand them. For example, The Sum or Difference Rule can be stated as "*The limit of a sum (difference) is the sum (difference) of the limits.*" Can you give a description for each property in your own words?

EX#1: Use the properties of limits to find each of the following limits. If a limit does not exist, state why.

A. $\lim_{x \rightarrow 3^+} [f(x) + g(x)]$

B. $\lim_{x \rightarrow 5} [3f(x) - 2g(x)]$

C. $\lim_{x \rightarrow -5^-} [f(x) - g(x)]$

D. $\lim_{x \rightarrow -7} \frac{3f(x)}{g(x)}$

E. $\lim_{x \rightarrow 3} 4[f(x) \cdot g(x)]$

F. $\lim_{x \rightarrow -3} [f(x)]^2$

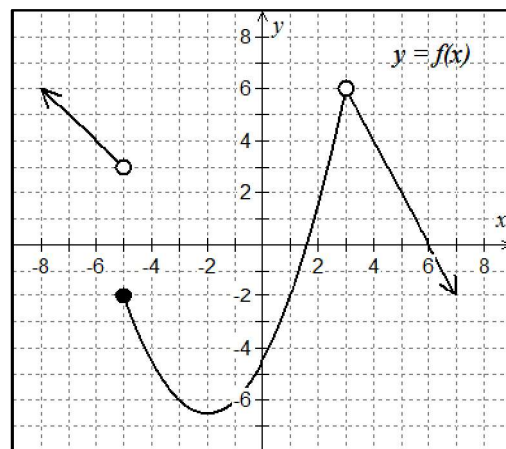
G. $\lim_{x \rightarrow 4} \sqrt{8g(x)}$

H. $\lim_{x \rightarrow -3} [f(x) + g(x)]$

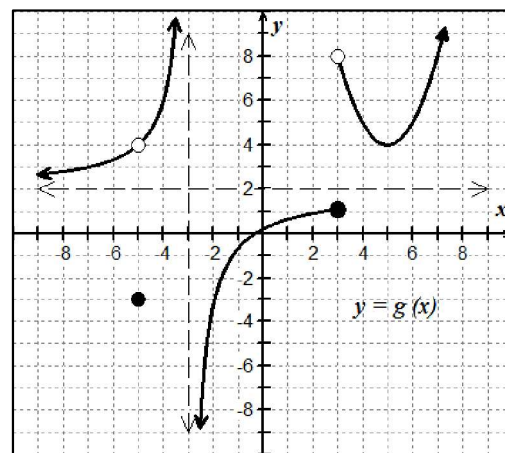
J. $\lim_{x \rightarrow 6} \frac{f(x)}{g(x)}$

K. $\lim_{x \rightarrow -2} f(g(x))$

Graph of $f(x)$



Graph of $g(x)$



EX #2: Find the limits if $\lim_{x \rightarrow c} f(x) = 3$, $\lim_{x \rightarrow c} g(x) = -2$, and $\lim_{x \rightarrow c} h(x) = 4$

A. $\lim_{x \rightarrow c} \sqrt{3h(x) - 2g(x)}$

B. $\lim_{x \rightarrow c} [f(x) \cdot 5g(x)]$

C. $\lim_{x \rightarrow c} [7 - g(x)]^2$

D. $\lim_{x \rightarrow c} \frac{2f(x) + 3h(x)}{h(x) - g(x)}$

E. $\lim_{x \rightarrow c} [h(x) \cdot (f(x) + 6)]$

F. $\lim_{x \rightarrow c} \frac{f(x)^2}{4 - g(x)}$

EX #3: Given $\lim_{x \rightarrow c} f(x) = 8$ and $\lim_{x \rightarrow c} g(x) = -4$, then find the following:

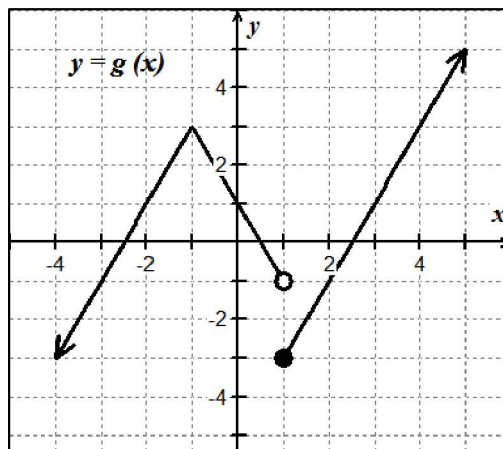
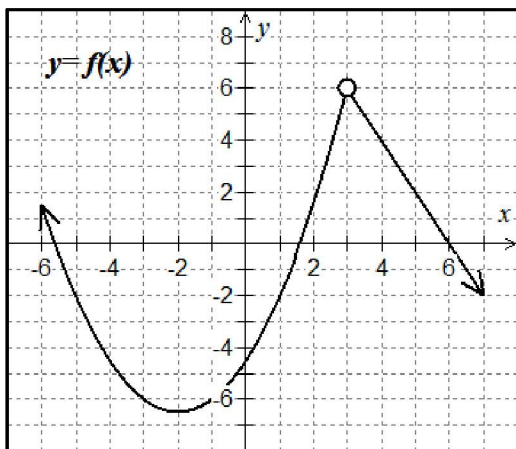
$$\lim_{x \rightarrow c} \left\{ 2f(x) \cdot g(x) - \sqrt[3]{f(x)} + 2[g(x)]^2 \right\}$$

EX #4: Find the limits of the following composition functions.

A. Given $f(x) = x^3 - x^2 + 4x - 1$ and $g(x) = -2x$, find $\lim_{x \rightarrow 2} f(g(x))$.

B. Given $f(x) = x^3 - x^2 + 4x - 1$ and $g(x) = -2x$, find $\lim_{x \rightarrow 2} g(f(x))$

EX #5: Use the graphs of $f(x)$ and $g(x)$ below to find the limits.



A. $\lim_{x \rightarrow 3} f(g(x))$

B. $\lim_{x \rightarrow -1} g(f(x))$

C. $\lim_{x \rightarrow 1} f(g(x))$

D. $\lim_{x \rightarrow -3} g(f(x))$